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| LAB 02 | |
|---|---|
| Aim: | Extraction of data using R |
| Theory: | Extraction of given rows and columns has always been one of the most important tasks which are especially required while working on data cleaning activities. In this article, we will be discussing all the sets of commands which are used to extract given rows and columns from a given dataframe in the R programming language.<br><br>❖ **Extraction of a single element from a dataframe:**<br>A single element can be extracted if the exact position of the item is known and is passed.<br>**Syntax**: *df [ row, column]*<br><br>❖ **extraction of more than one rows with all columns:**<br><br>c in c() means 'combine'. c function is a generic function which combines it arguments to form vectors. When row numbers to be extracted are specified and column number is not specified, selected rows with all columns are displayed.<br><br>**Syntax:** *df[ c(n,m), ]*<br><br>❖ **extraction of range of rows and columns:**<br><br>Supply rows and columns with ranges.<br><br>**Syntax:** *df[p:q, n:m]* |
| Code: | vsda-2-64.r<br>Source on Save Run Source<br>`df <- data.frame(team=c("sheikh","tanmay","tejas","parth","omkar"),`<br>`marks=c(99,90,86,88,95),`<br>`cgpa=c(9,8,8,7,5),`<br>`rank=c(1,2,2,3,4))`<br>`df`<br>`df[2,]`<br>`df[c(2,4,5),]`<br>`df[1:3]`<br>`df[df$marks>90,]`<br>`df[df$marks>89&df$cgpa>7,]` |
| Output: | |

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```
> df
     team marks cgpa rank
1  sheikh    99    9    1
2  tanmay    90    8    2
3   tejas    86    8    2
4   parth    88    7    3
5   omkar    95    5    4
> df[2,]
     team marks cgpa rank
2  tanmay    90    8    2
> df[c(2,4,5),]
     team marks cgpa rank
2  tanmay    90    8    2
4   parth    88    7    3
5   omkar    95    5    4
> df[1:3]
     team marks cgpa
1  sheikh    99    9
2  tanmay    90    8
3   tejas    86    8
4   parth    88    7
```

```
> df[df$marks>90,]
     team marks cgpa rank
1  sheikh    99    9    1
5   omkar    95    5    4
> df[df$marks>89&df$cgpa>7,]
     team marks cgpa rank
1  sheikh    99    9    1
2  tanmay    90    8    2
>
```

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| Conclusion | Extraction of data using R executed successfully. |
| Name Of Faculty: | Prof. Akhil Jajulwar |